

NIHAL POTHUNOORI

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EDUCATION

University of Illinois at Urbana-Champaign

Expected May 2027

B.S. in Computer Engineering

Relevant Coursework: Control Systems; Deep Learning; Principles of Safe Autonomy; Analog Signal Processing; Computer Systems & Programming

SKILLS

Programming: C++, Python (PyTorch, OpenCV, NumPy), CUDA, Matlab, Java, C#/.NET

ML & Robotics: ROS2, SLAM, Sensor Fusion, State Estimation, Deep/Reinforcement Learning, Diffusion Models, Vision-Language-Action Models, Computer Vision, Simulation (Gazebo, Isaac Sim/Lab)

Tools: Linux, Git, Docker, Kubernetes, AWS; Design: CAD (Fusion360), KiCAD, Altium, PCB Design (I2C, SPI)

EXPERIENCE

Rivian

Jun 2026 – Present

Machine Learning (Research) Intern

Champaign, IL

- Researching world models and VLAs for autonomous driving on Rivian's custom RAP1 hardware; optimizing inference via INT4/INT8 quantization, structured pruning, and custom CUDA/Triton kernels for fused attention, reducing latency by **3.2×**.
- Integrating model-based RL into the world model training loop to improve long-horizon vehicle dynamics prediction; compiled subgraphs to TensorRT, hand-tuning tile sizes and memory layouts for RAP1's cache hierarchy to hit real-time thresholds.

DASLABS, University of Illinois

Jan 2026 – Present

Undergraduate Researcher

Champaign, IL

- Designing a VLA architecture in Python/PyTorch for a hybrid rigid-soft arm that perceives environmental geometry to identify viable anchor and bracing structures for the soft link; optimizing manipulation efficiency beyond passive obstacle avoidance.
- Built a depth-camera pipeline to isolate the deformable soft-arm backbone; performed real-time 3D point-cloud processing and registration to reconstruct arm geometry in world coordinates, and applied SAM3-based binary mask segmentation for online policy evaluation against ground-truth pose benchmarks.

College of Education, University of Illinois

Aug 2025 – Present

Web Developer/IT Support

Champaign, IL

- Built an automated enterprise-level permissions management system for ~2,500 users via a high-performance .NET 8 C# library integrated with Azure Active Directory to streamline identity and access management across the department.
- Developing an intuitive web-based interface in HTML to enable non-technical administrators to execute complex operations on user roles, reducing manual IT support tickets and ensuring 99.9% consistency in organizational permission hierarchies.

Autonomous Materials Systems Group, University of Illinois

Aug 2025 – Jan 2026

Undergraduate Researcher

Champaign, IL

- Designed a 3-axis robotic manipulator using inverse kinematics and automated pipette dispenser for the preparation of DCPD polymer samples within a DARPA-funded laboratory focused on rapid, autonomous material synthesis and chemical discovery.
- Architected and deployed a PyTorch-based reinforcement learning framework with automated data pipelines for closed-loop process control, optimizing between environmental variables and mechanical actuation to achieve a 75% reduction in total sample preparation time.

PROJECTS

Soccer Counterfactual Video Generation

Apr 2026

- Implemented a text-conditioned video generation model from scratch in Python/PyTorch inspired by the Wan paper; architected a full VAE + Diffusion Transformer (DiT) pipeline capable of generating counterfactual simulation scenarios from natural language prompts, with frames decoded simultaneously from a learned continuous latent space.
- Built text conditioning via a T5 encoder into DiT cross-attention (K/V); applied AdaLN timestep embeddings and classifier-free guidance for controllable inference; trained end-to-end on ~20,000 5-second clips from SoccerNetv2.

Laundry Robot

Jan 2026

- Built a full AV-style perception and autonomy stack in ROS2: deployed a YOLOv8 segmentation model for real-time object detection, then back-projected 2D detections into 3D world coordinates via a custom vision-to-world service.
- Implemented an Extended Kalman Filter (EKF) to fuse asynchronous odometry and visual observations for continuous vehicle localization and state estimation; EKF pose estimates drove a Stanley Controller for precise closed-loop path following.

Graph Neural Network for School Zone Detection – Hackathon

Apr 2026

- Framed road-network attribute prediction as a graph learning problem: nodes encode road segments with geographic features (speed limits, road signs, landmarks); edges encode spatial adjacency. Trained a GNN from scratch in PyTorch on Iowa map data to propagate school-zone context across the road graph.
- Model predicts speed limits for all road segments surrounding a known school-zone anchor, achieving 99.7% accuracy within ± 5 mph and 86.7% within ± 2 mph; outperformed all other teams at the hackathon.